KCL AND KVL

CIRCUIT ANALYSIS METHOD

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

LABORATORY

50%

EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Labeling Variables

Current Flow Convention

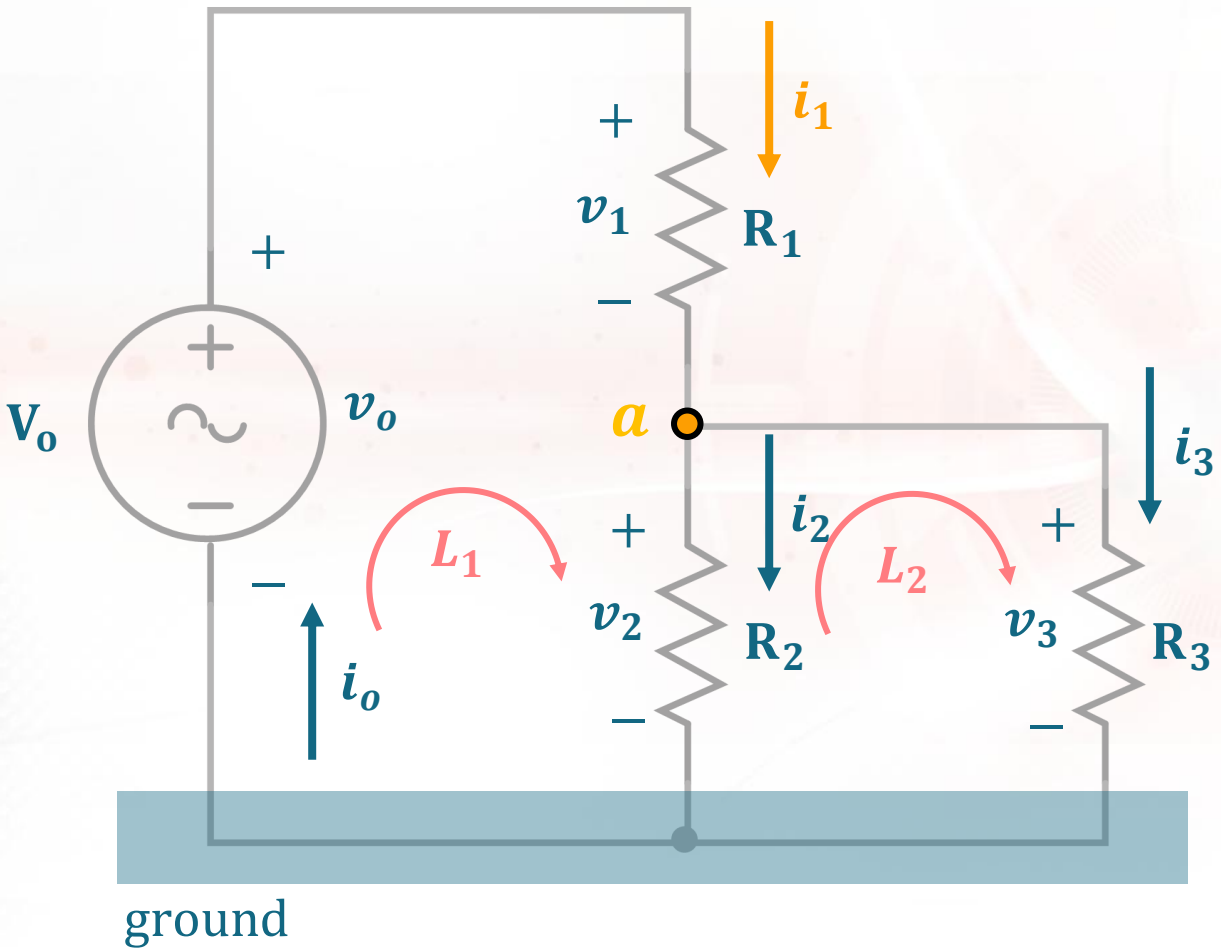
Voltage Loop Convention

Kirchhoff's Current Law (KCL)

Kirchhoff's Voltage Law (KVL)

CIRCUIT CONVENTION

LABELING VARIABLES



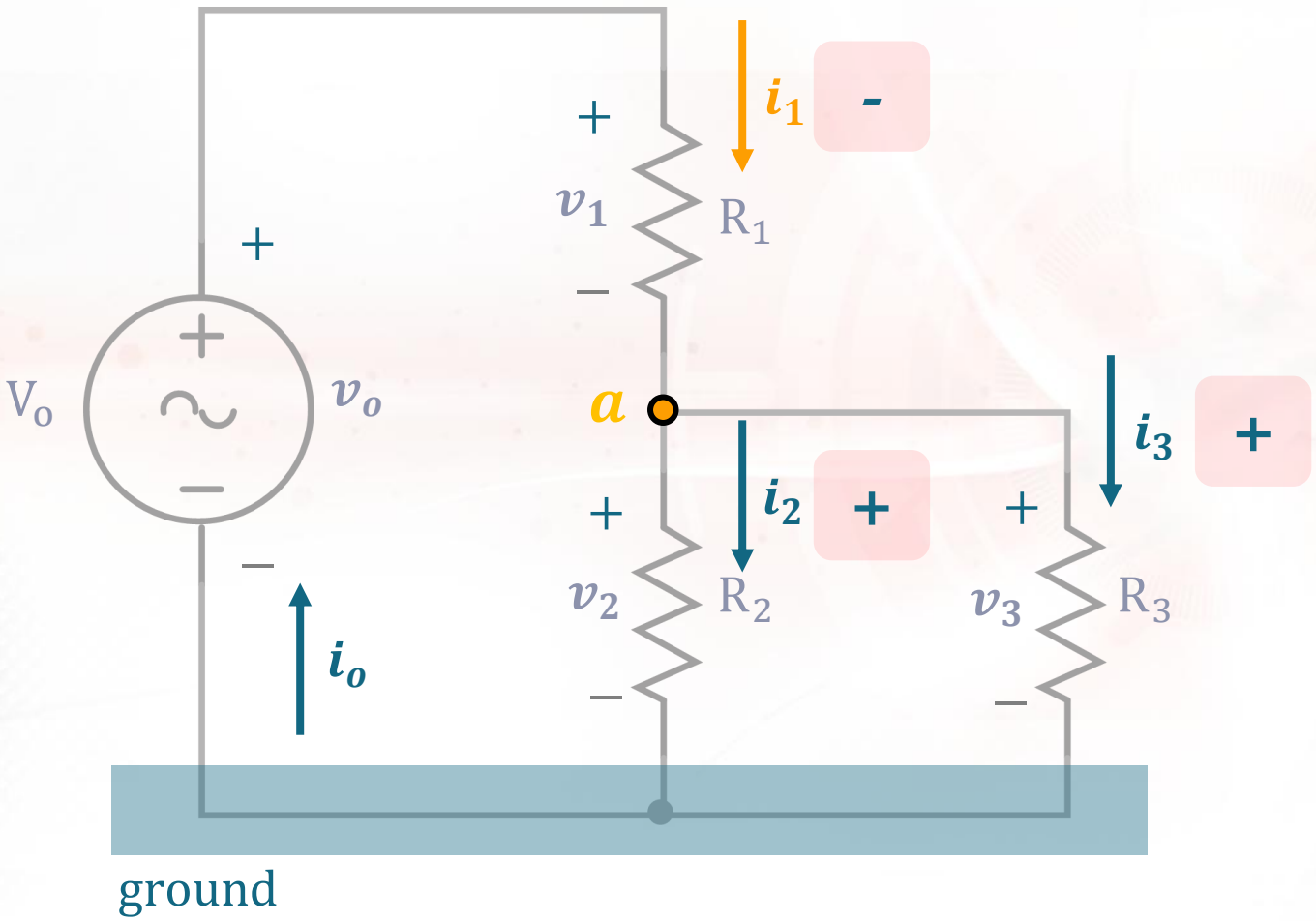
Select **reference node** (⏚) from which voltages are measured

Label **voltages** with respect to reference node

Label **current** flowing into the positive (+) terminal of the element

Create a **voltage loop**

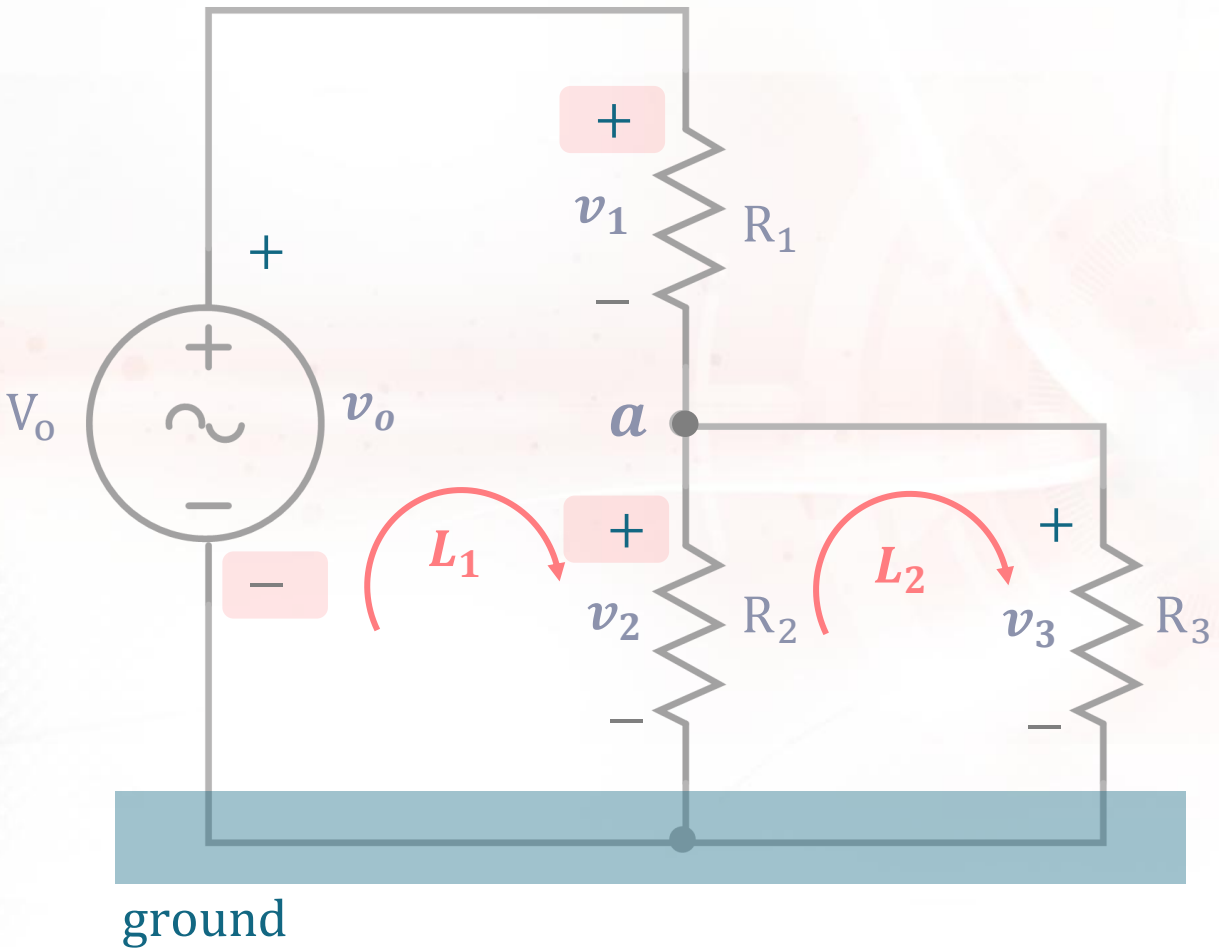
CURRENT FLOW CONVENTION



Current is taken to be **negative** going **in**to the node

Current is taken to be **positive** going **out** from the node

VOLTAGE LOOP CONVENTION



$@L_1,$

$-v_0$

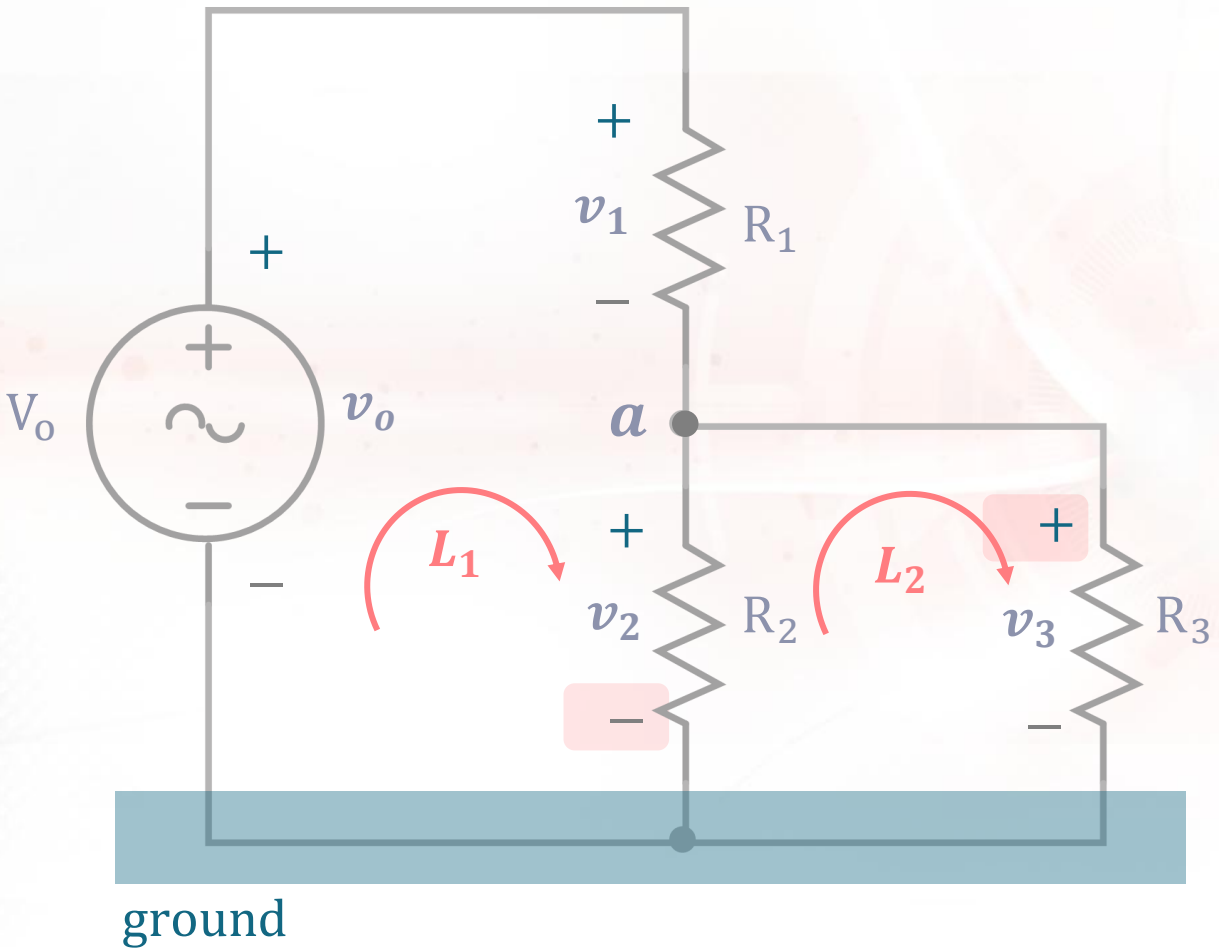
$+v_1$

$+v_2$

$@L_2,$

The "**sign**" of voltage of the element is the **first** sign the loop encounters

VOLTAGE LOOP CONVENTION



$@L_1,$

$-v_0$

$+v_1$

$+v_2$

$@L_2,$

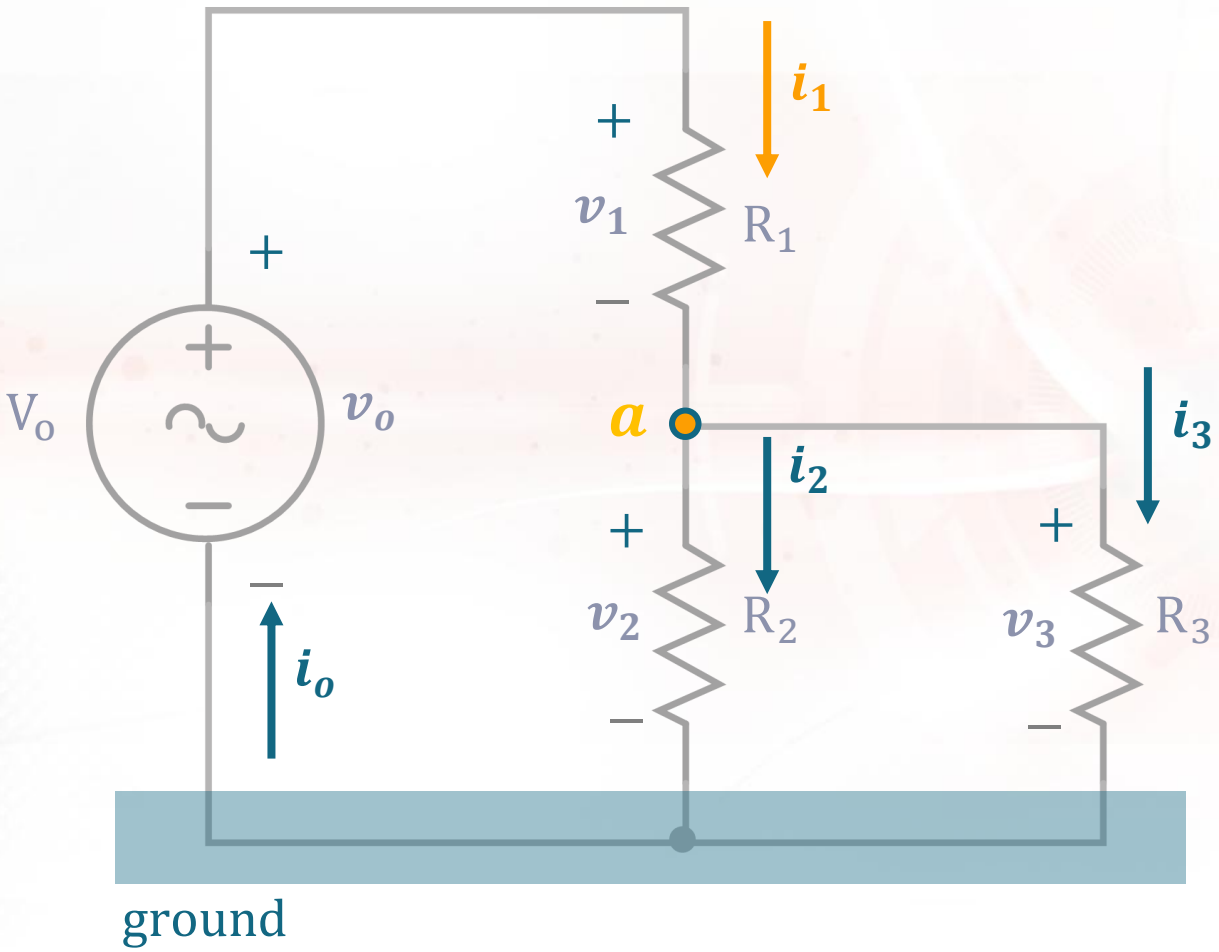
$-v_2$

$+v_3$

The "**sign**" of voltage of the element is the **first** sign the loop encounters

KCL AND KVL

KIRCHHOFF'S CURRENT LAW



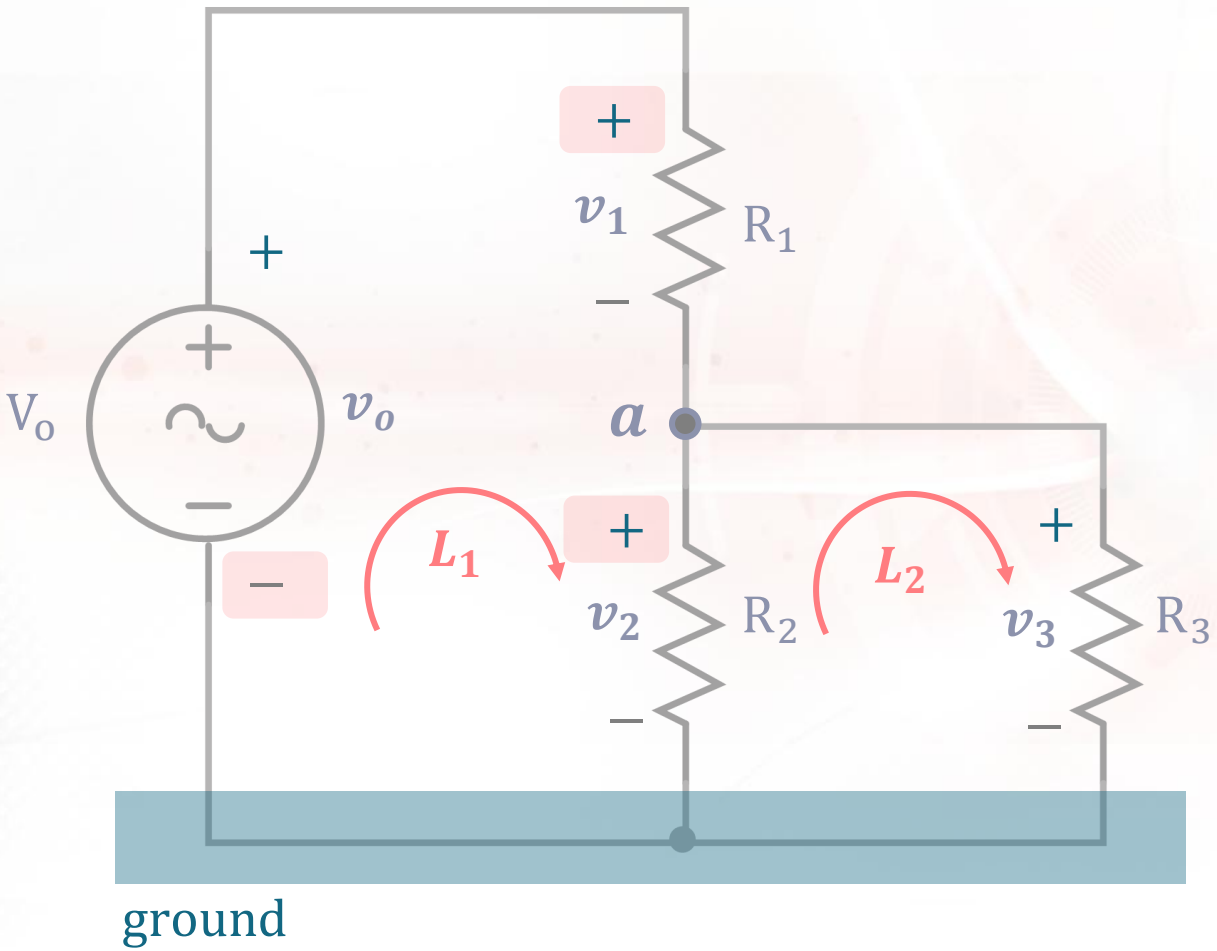
The summation of **currents** going-in and going-out a node is **zero**

$$\sum_{node} i_j = 0$$

KCL @ a,

$$-i_1 + i_2 + i_3 = 0$$

KIRCHHOFF'S VOLTAGE LAW



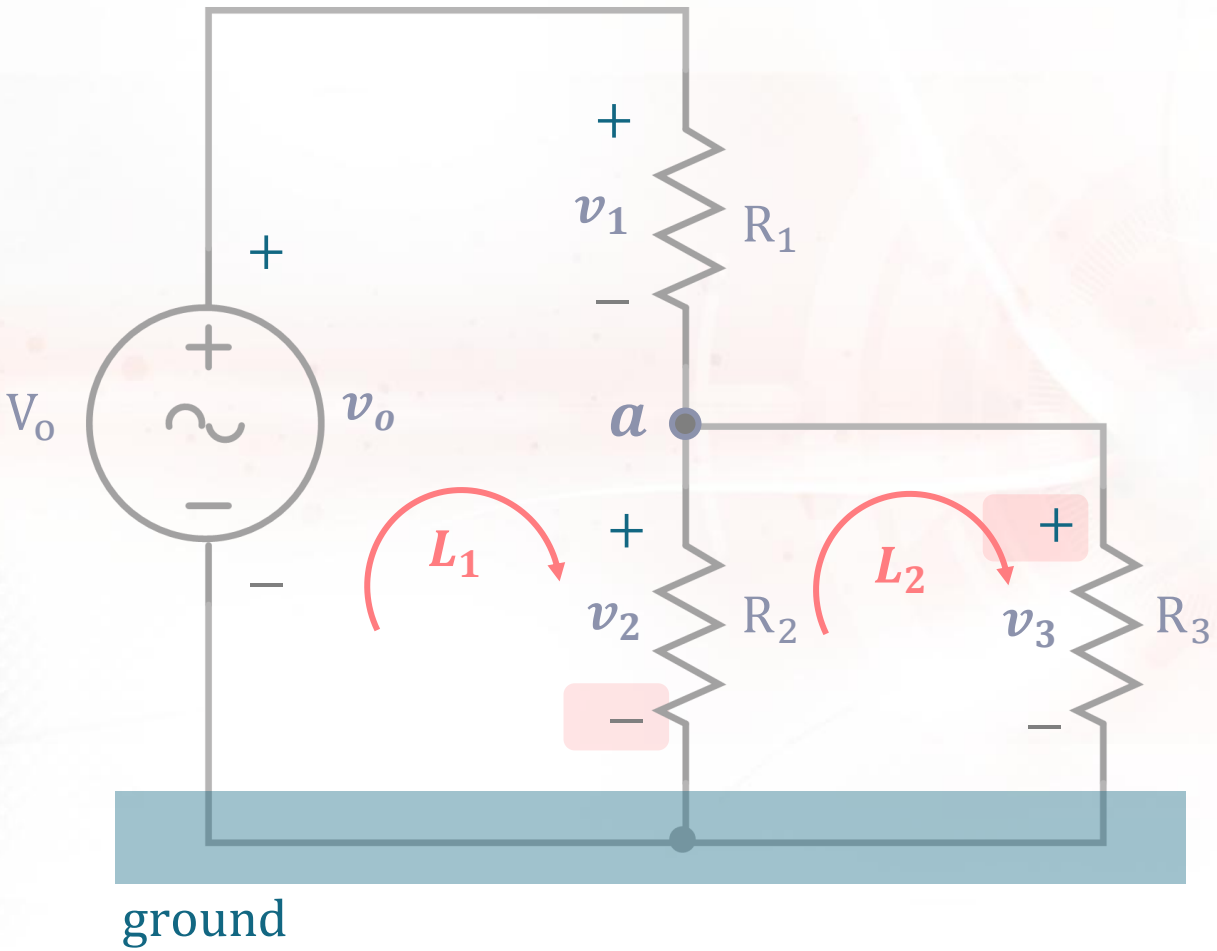
The summation of **voltages** in a closed-loop is **zero**

$$\sum_{loop} v_j = 0$$

KVL @$\boldsymbol{L_1}$,

$$-v_o + v_1 + v_2 = 0$$

KIRCHHOFF'S VOLTAGE LAW



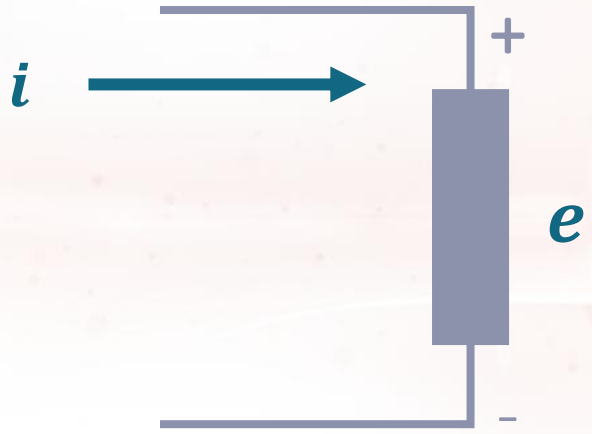
The summation of **voltages** in a closed-loop is **zero**

$$\sum_{loop} v_j = 0$$

KVL @$\boldsymbol{L_2}$,

$$-v_2 + v_3 = 0$$

POWER



The power consumed by element, e

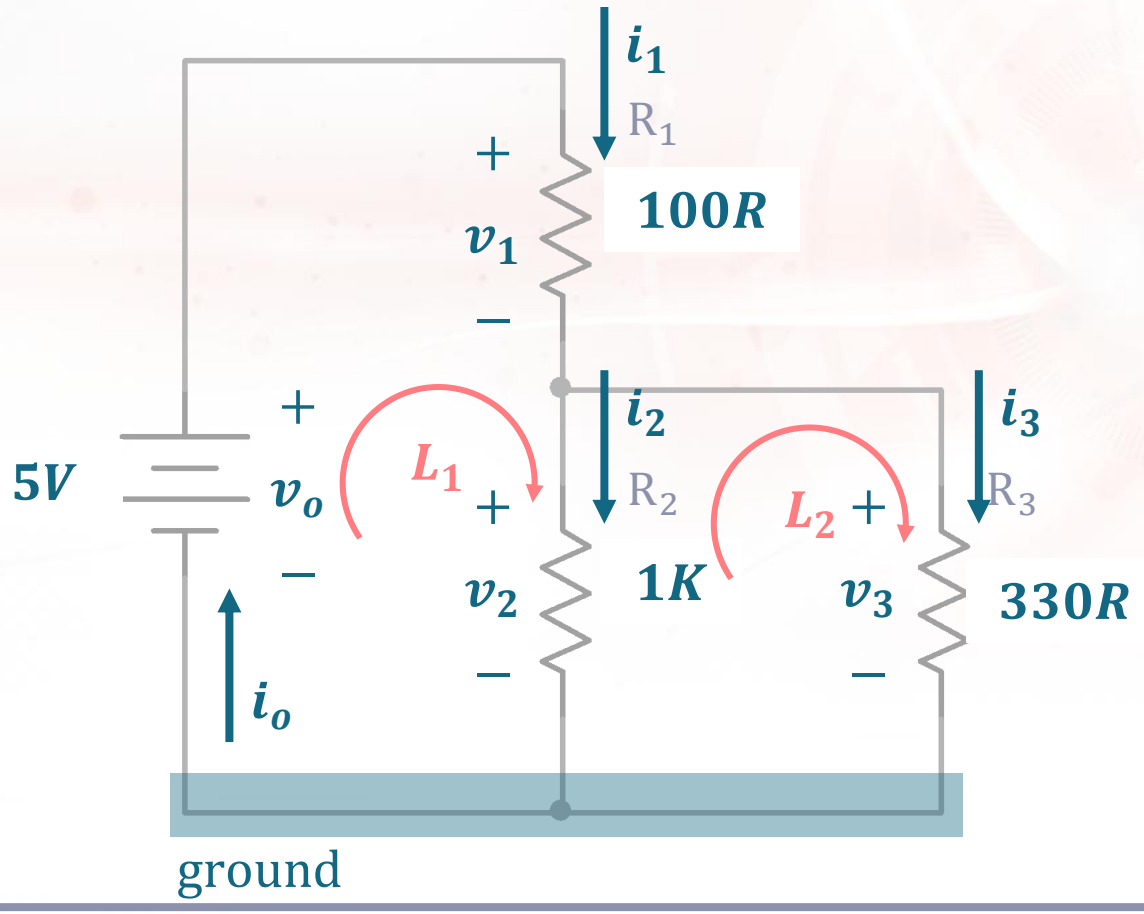
$$P = vi$$

$$P = i^2 R$$

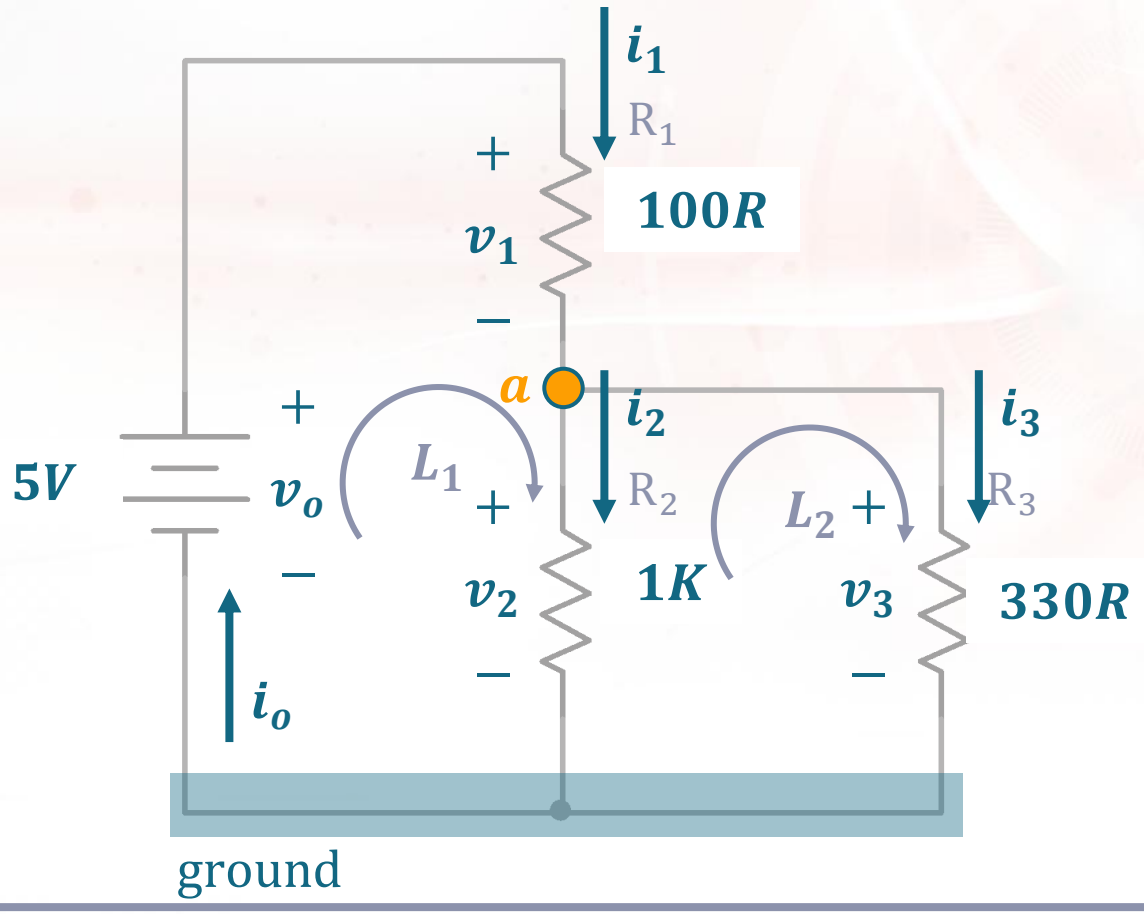
$$P = \frac{v^2}{R}$$

Unit: Watt (W)

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



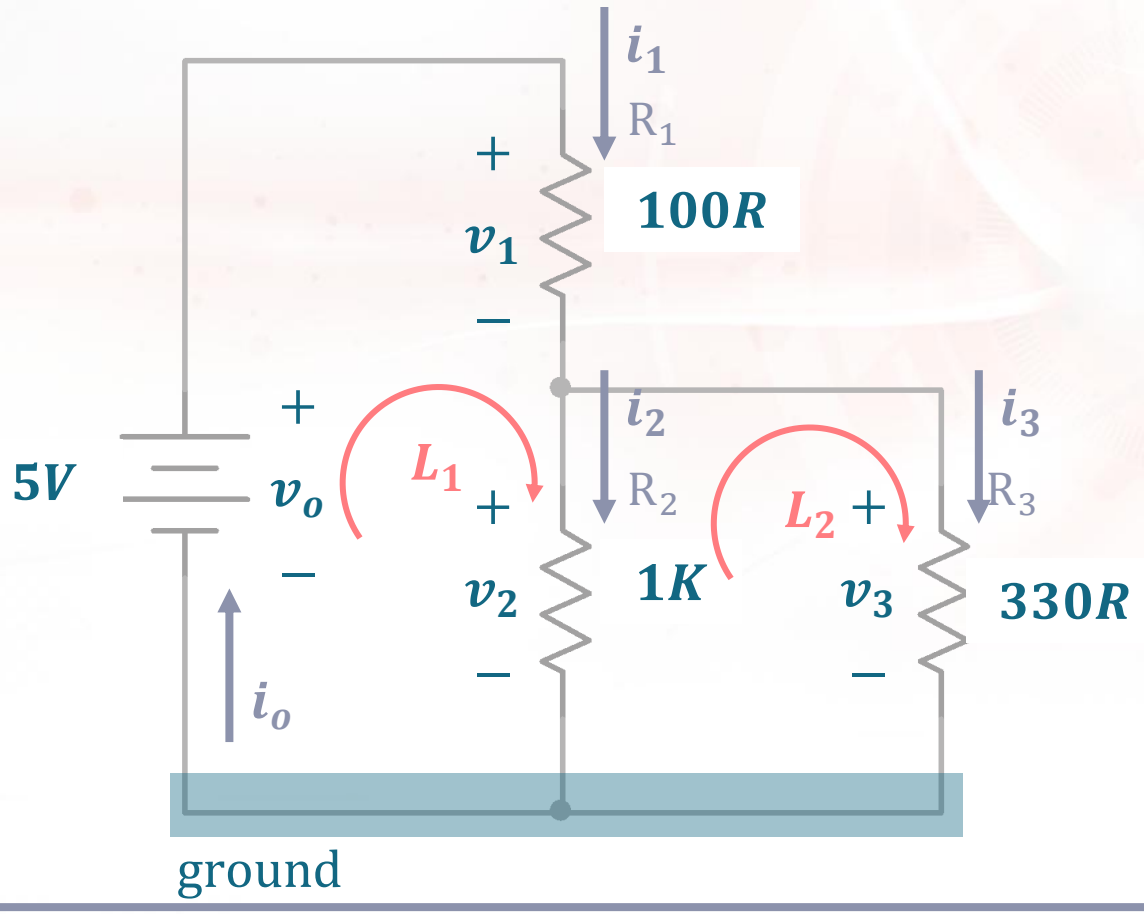
KCL @ a

$$-i_1 + i_2 + i_3 = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = 0$$

$$-\frac{v_1}{100R} + \frac{v_2}{1K} + \frac{v_3}{330R} = 0$$ equation 1

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



KVL @ L_1

$$-v_o + v_1 + v_2 = 0$$

$$-5 + v_1 + v_2 = 0$$

$$v_1 + v_2 = 5 \quad \text{equation 2}$$

KVL @ L_2,

$$-v_2 + v_3 = 0 \quad \text{equation 3}$$

Gaussian Elimination,

Augmented Matrix

$$\begin{array}{cccc} v_1 & v_2 & v_3 & \\ \end{array}$$

$$\left|\begin{array}{cccc} -\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\ 1 & 1 & 0 & 5 \\ 0 & -1 & 1 & 0 \end{array}\right|$$

echelon form

$$-\frac{v_1}{100R} + \frac{v_2}{1K} + \frac{v_3}{330R} = 0 \quad (1)$$

$$v_1 + v_2 = 5 \quad (2)$$

$$-v_2 + v_3 = 0 \quad (3)$$

Gaussian Elimination,

Augmented Matrix

$$
\begin{array}{c}
\begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\left|\begin{array}{cccc}
-\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\
1 & 1 & 0 & 5 \\
0 & -1 & 1 & 0
\end{array}\right|
\end{array}
\begin{array}{l}
r_1 \\
r_2 \\
r_3
\end{array}
\quad \longleftarrow \quad 100r_1 + r_2
$$

echelon form

Gaussian Elimination,

Augmented Matrix

$$
\begin{array}{c}
\begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\left|\begin{array}{cccc}
-\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\
0 & 1 & 0 & 5 \\
0 & -1 & 1 & 0
\end{array}\right|
\end{array}
\begin{array}{l}
r_1 \\
r_2 \\
r_3
\end{array}
\quad \longleftarrow \quad 100r_1 + r_2
$$

echelon form

Gaussian Elimination,

Augmented Matrix

$$
\begin{array}{c}
\begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\left|\begin{array}{cccc}
-\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\
0 & 1.1 & 0 & 5 \\
0 & -1 & 1 & 0
\end{array}\right|
\end{array}
\begin{array}{l}
r_1 \\ r_2 \\ r_3
\end{array}
\quad \longleftarrow \quad 100r_1 + r_2
$$

echelon form

Gaussian Elimination,

Augmented Matrix

$$
\begin{array}{c}
\begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\left|\begin{array}{cccc}
-\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\
0 & 1.1 & \frac{10}{33} & 5 \\
0 & -1 & 1 & 0
\end{array}\right|
\end{array}
\begin{array}{l}
r_1 \\ r_2 \\ r_3
\end{array}
\quad \longleftarrow \quad 100r_1 + r_2
$$

echelon form

Gaussian Elimination,

Augmented Matrix

$$
\begin{array}{c}
\begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\left|\begin{array}{cccc}
-\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\
0 & 1.1 & \frac{10}{33} & 5 \\
0 & -1 & 1 & 0
\end{array}\right|
\end{array}
\begin{array}{l}
r_1 \\ r_2 \\ r_3 \leftarrow 1.1r_3 + r_2
\end{array}
$$

echelon form

Gaussian Elimination,

Augmented Matrix

$$
\begin{array}{c}
\begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\left|\begin{array}{cccc}
-\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\
0 & 1.1 & \frac{10}{33} & 5 \\
0 & 0 & 1 & 0
\end{array}\right|
\end{array}
\begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}
\longleftarrow 1.1r_3 + r_2
$$

echelon form

Gaussian Elimination,

Augmented Matrix

$$
\begin{array}{c}
\begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\left|\begin{array}{cccc}
-\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\
0 & 1.1 & \frac{10}{33} & 5 \\
0 & 0 & 1.4 & 0
\end{array}\right|
\end{array}
\begin{array}{l}
r_1 \\ r_2 \\ r_3 \leftarrow 1.1r_3 + r_2
\end{array}
$$

echelon form

Gaussian Elimination,

Augmented Matrix

$$
\begin{array}{c}
\begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\left|\begin{array}{cccc}
-\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\
0 & 1.1 & \frac{10}{33} & 5 \\
0 & 0 & 1.4 & 5
\end{array}\right|
\end{array}
\begin{array}{l}
r_1 \\ r_2 \\ r_3 \leftarrow 1.1r_3 + r_2
\end{array}
$$

echelon form

Gaussian Elimination,

Augmented Matrix

$$
\begin{array}{cccc}
v_1 & v_2 & v_3 & \\
\end{array}
$$

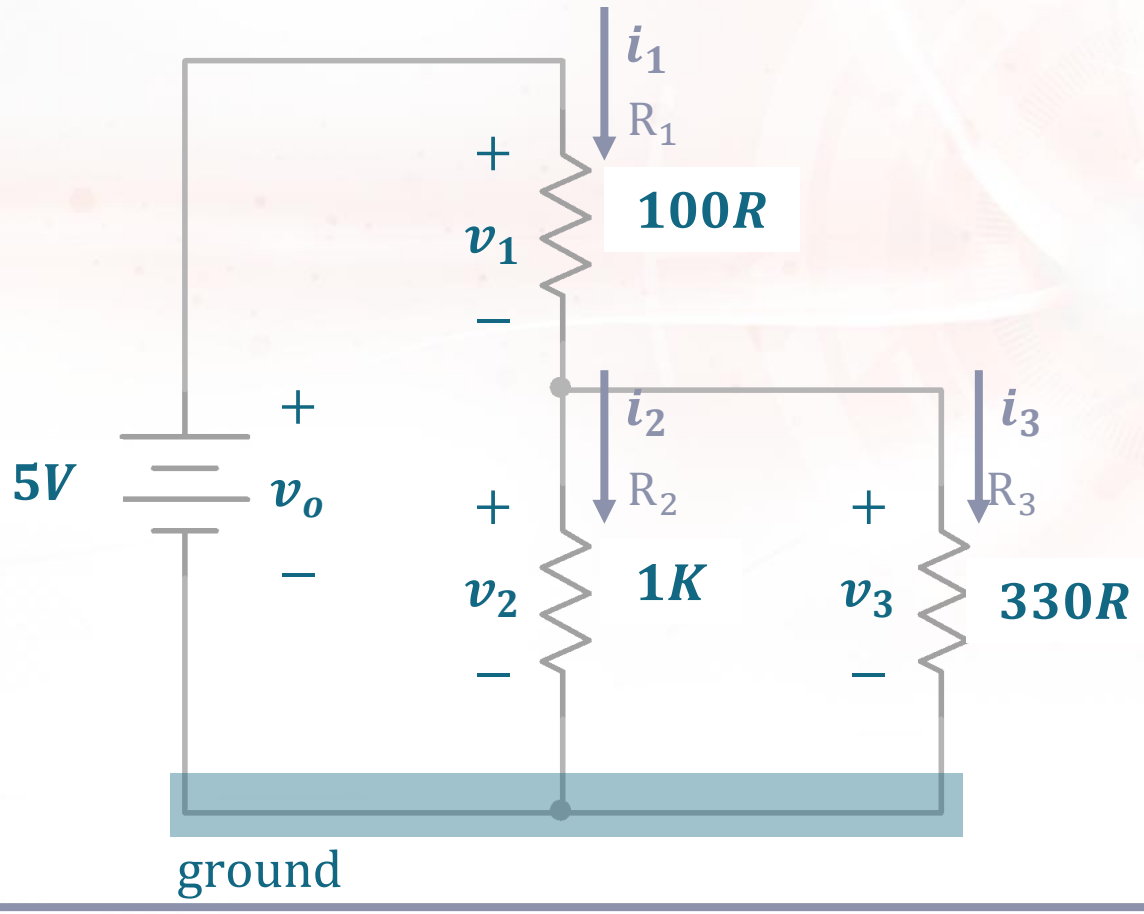
$$
\left|\begin{array}{cccc}
-\frac{1}{100} & \frac{1}{1000} & \frac{1}{330} & 0 \\
0 & 1.1 & \frac{10}{33} & 5 \\
0 & 0 & 1.4 & 5
\end{array}\right|
\begin{array}{l}
r_1 \\ r_2 \\ r_3
\end{array}
\qquad
\begin{array}{l}
-\frac{v_1}{100R} + \frac{v_2}{1K} + \frac{v_3}{330R} = 0 \\
1.1v_2 + \frac{10}{33}v_3 = 5 \\
1.4v_3 = 5
\end{array}
$$

echelon form

$$\boxed{v_3 = 3.57\,V}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$$-\frac{v_1}{100R}+\frac{v_2}{1K}+\frac{v_3}{330R}=0$$

$$1.1v_2+\frac{10}{33}v_3=5$$

$$1.4v_3=5$$

$$\boxed{v_3=3.57\,V}$$

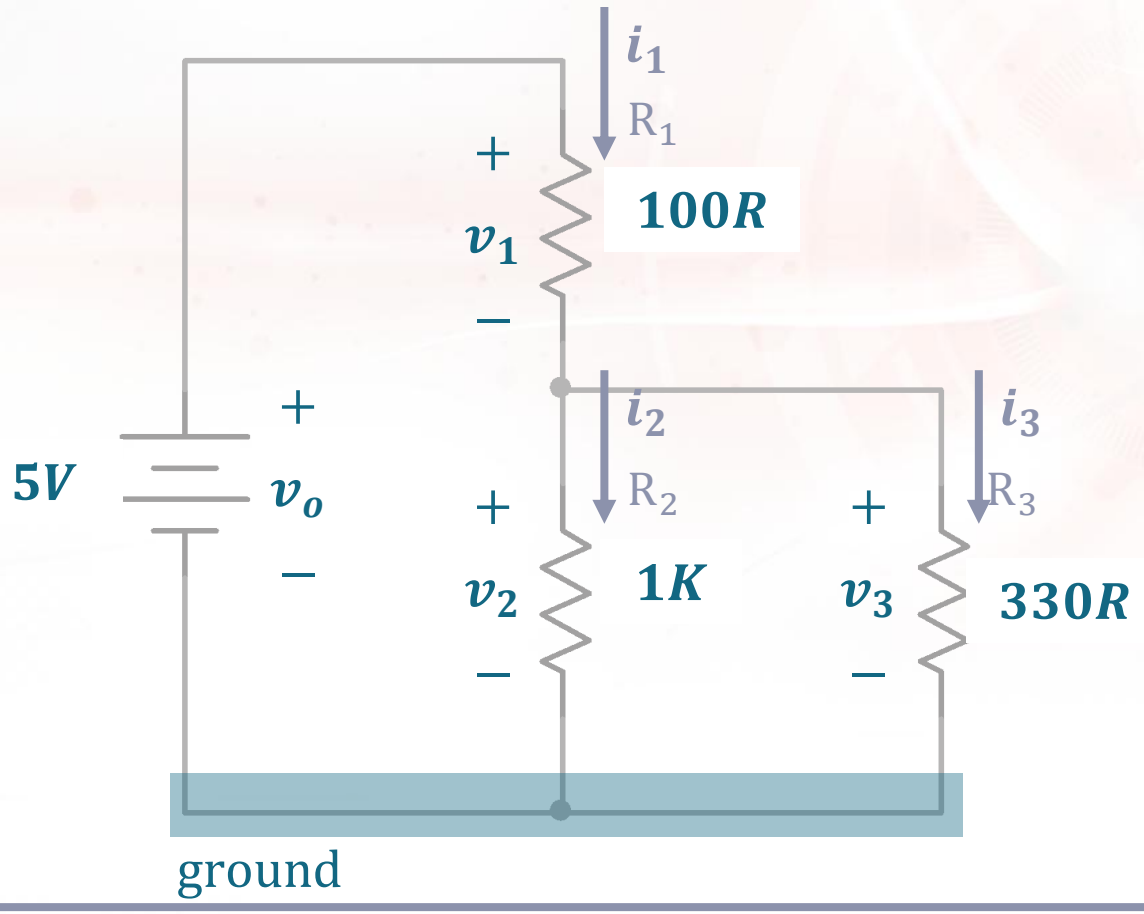
ans

$$-v_2+v_3=0 \quad (3)$$

$$\boxed{v_2=3.57\,V}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$$-\frac{v_1}{100R}+\frac{v_2}{1K}+\frac{v_3}{330R}=0$$

$$-\frac{v_1}{100R}+\frac{3.57}{1K}+\frac{3.57}{330R}=0$$

$$-\frac{v_1}{100R}+0.01439=0$$

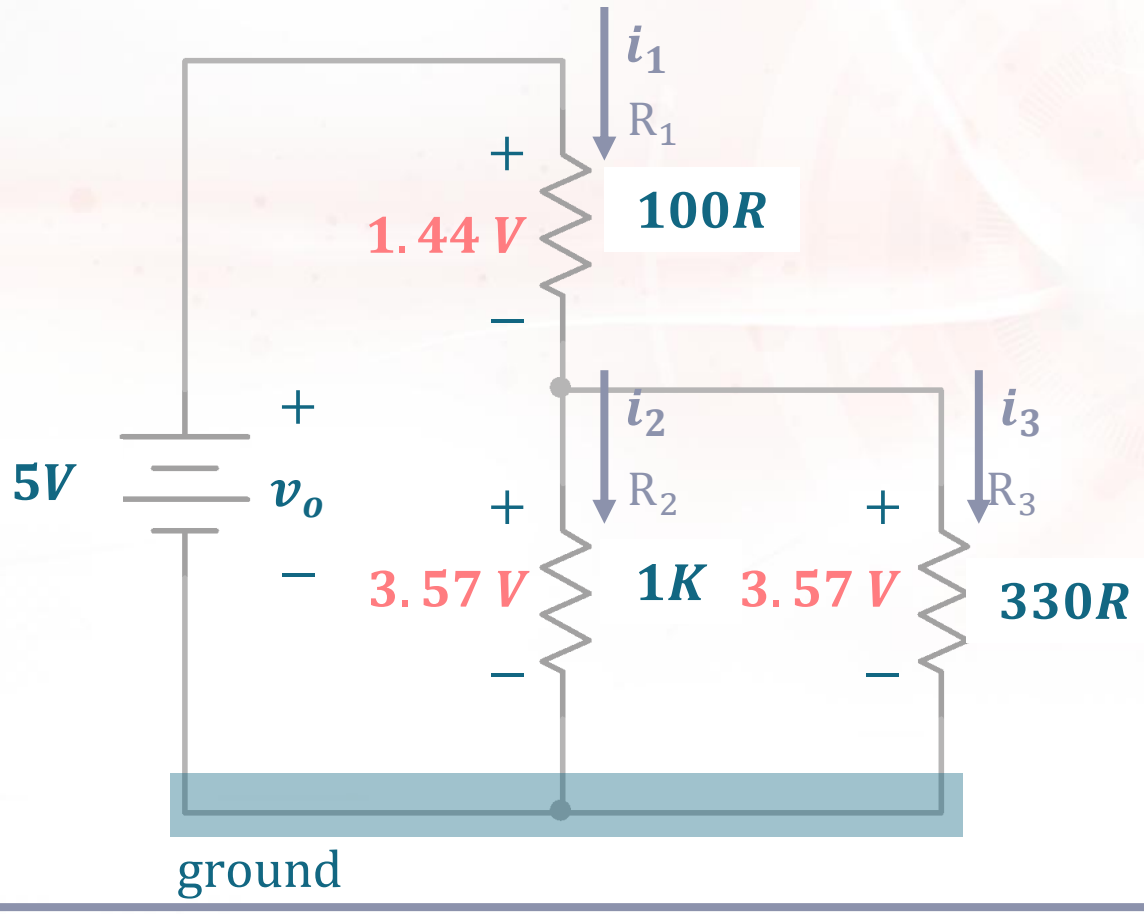
$$\frac{v_1}{100R}=0.01439$$

$$v_1=(100)0.01439$$

$$\boxed{v_1=1.44\text{ V}}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$$i_1 = \frac{v_1}{R_1}$$

$$i_1 = \frac{1.44}{100}$$

$$\boxed{i_1 = 14.4\ mA}$$

ans

$$i_2 = \frac{v_2}{R_2}$$

$$i_2 = \frac{3.57}{1K}$$

$$\boxed{i_2 = 3.57\ mA}$$

ans

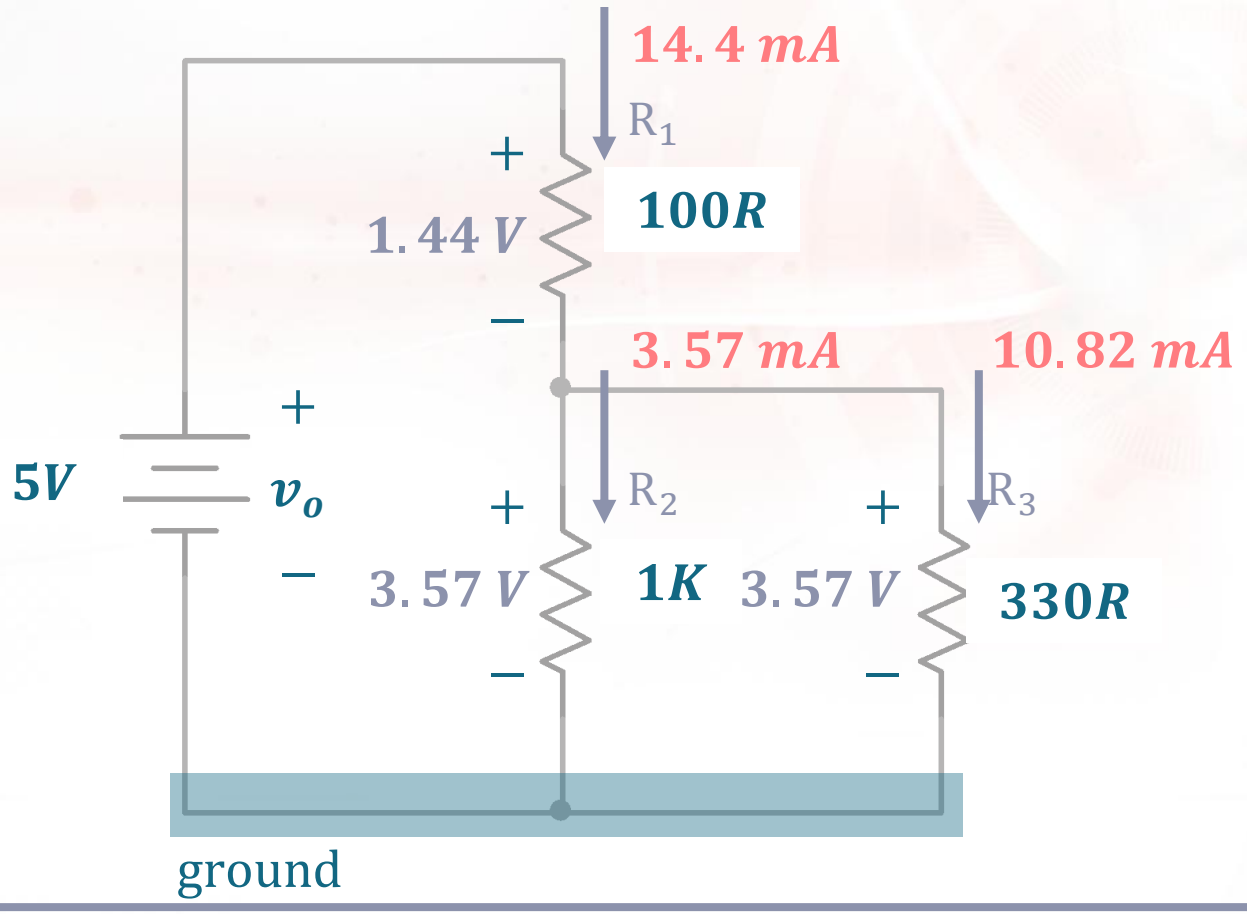
$$i_3 = \frac{v_3}{R_3}$$

$$i_3 = \frac{3.57}{330}$$

$$\boxed{i_3 = 10.82\ mA}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$$P_1 = v_1 i_1$$

$$P_1 = (1.44)14.4\ mA$$

$$\boxed{P_1 = 20.74\ mW}$$

ans

$$P_2 = v_2 i_2$$

$$P_2 = (3.57)3.57\ mA$$

$$\boxed{P_2 = 12.74\ mW}$$

ans

$$P_3 = v_3 i_3$$

$$P_3 = (3.57)10.82\ mA$$

$$\boxed{P_3 = 38.63\ mW}$$

ans

LABORATORY